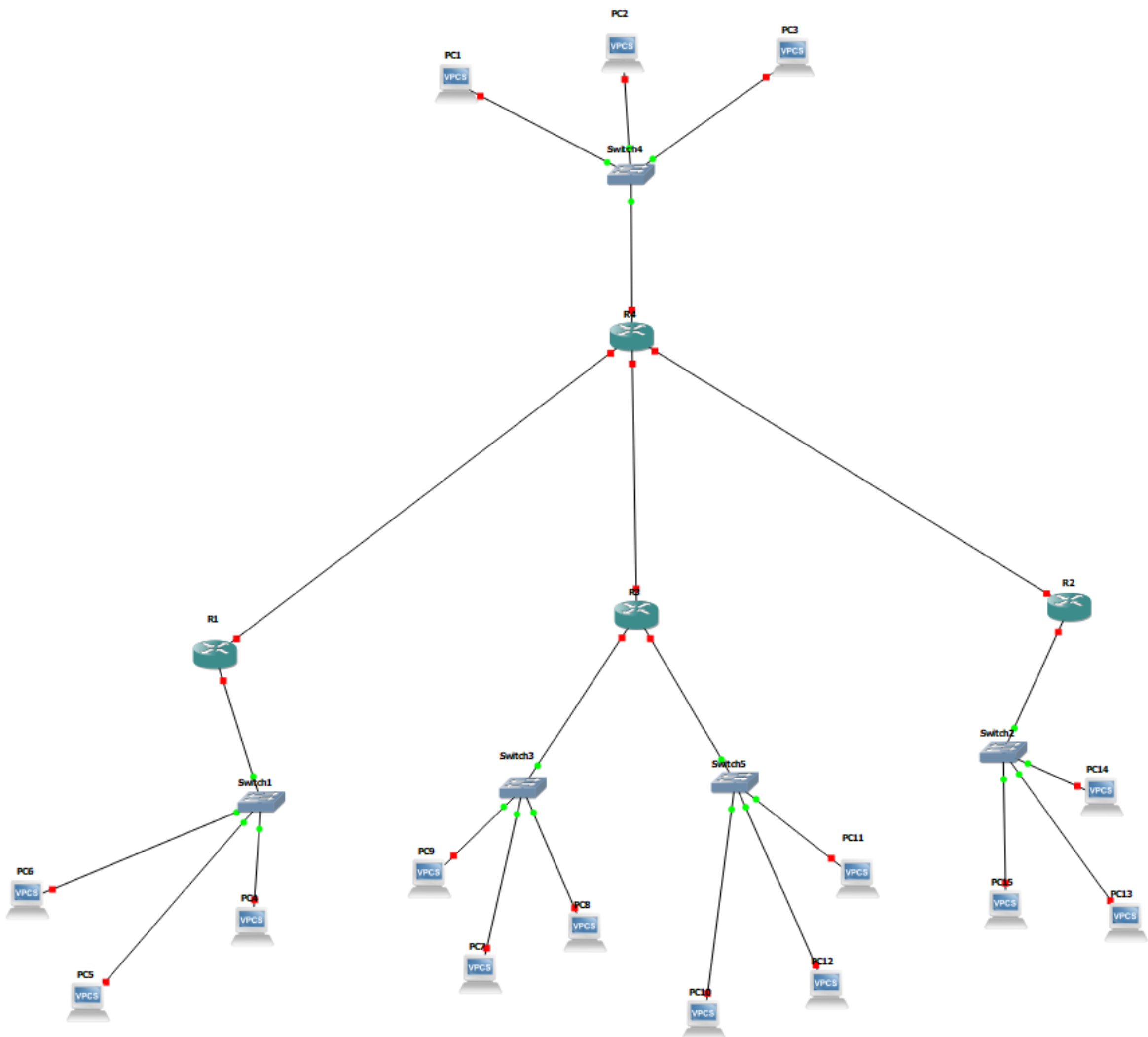
Part 1:-


PC1
PC2
PC3
Switch4
R1
R2
Switch1
Switch3
Switch5
Switch2
PC6
PC5
PC9
PC8
PC11
PC12
PC14
PC13
VPCS

Configuring IP addresses for each router interface with the corresponding subnet.

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.0.6     YES NVRAM  up                    up
FastEthernet0/1        192.168.0.206   YES NVRAM  up                    up
FastEthernet1/0        unassigned      YES NVRAM  administratively down down
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
R1#
```

```
R3#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.0.14    YES NVRAM  up                    up
FastEthernet0/1        192.168.0.198   YES NVRAM  up                    up
FastEthernet1/0        192.168.0.22    YES NVRAM  up                    up
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
R3#
```

```
R2#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.0.30    YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES NVRAM  administratively down down
FastEthernet1/0        192.168.0.190   YES NVRAM  up                    up
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
```

```
R4#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.0.205   YES NVRAM  up                    up
FastEthernet0/1        192.168.0.197   YES NVRAM  up                    up
FastEthernet1/0        192.168.0.189   YES NVRAM  up                    up
FastEthernet2/0        192.168.0.46    YES NVRAM  up                    up
```

Configuring RIP routing for each router based on connected IPs

```
R1#show ip RIP database
192.168.0.0/24    auto-summary
192.168.0.0/29    directly connected, FastEthernet0/0
192.168.0.8/29
    [2] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.16/29
    [2] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.24/29
    [2] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.40/29
    [1] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.184/29
    [1] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.192/29
    [1] via 192.168.0.205, 00:00:26, FastEthernet0/1
192.168.0.200/29    directly connected, FastEthernet0/1
R1#
```

```
R3#sh ip rip database
192.168.0.0/24    auto-summary
192.168.0.0/29
    [2] via 192.168.0.197, 00:00:15, FastEthernet0/1
192.168.0.8/29    directly connected, FastEthernet0/0
192.168.0.16/29    directly connected, FastEthernet1/0
192.168.0.24/29
    [2] via 192.168.0.197, 00:00:15, FastEthernet0/1
192.168.0.40/29
    [1] via 192.168.0.197, 00:00:15, FastEthernet0/1
192.168.0.184/29
    [1] via 192.168.0.197, 00:00:15, FastEthernet0/1
192.168.0.192/29    directly connected, FastEthernet0/1
192.168.0.200/29
    [1] via 192.168.0.197, 00:00:15, FastEthernet0/1
R3#
```

```
R2#sh ip rip database
192.168.0.0/24    auto-summary
192.168.0.0/29
    [2] via 192.168.0.189, 00:00:26, FastEthernet1/0
192.168.0.8/29
    [2] via 192.168.0.189, 00:00:26, FastEthernet1/0
192.168.0.16/29
    [2] via 192.168.0.189, 00:00:26, FastEthernet1/0
192.168.0.24/29    directly connected, FastEthernet0/0
192.168.0.40/29
    [1] via 192.168.0.189, 00:00:26, FastEthernet1/0
192.168.0.184/29    directly connected, FastEthernet1/0
192.168.0.192/29
    [1] via 192.168.0.189, 00:00:26, FastEthernet1/0
192.168.0.200/29
    [1] via 192.168.0.189, 00:00:26, FastEthernet1/0
R2#
```

```
R4#sh ip rip database
192.168.0.0/24    auto-summary
192.168.0.0/29
    [1] via 192.168.0.206, 00:00:29, FastEthernet0/0
192.168.0.8/29
    [1] via 192.168.0.198, 00:00:01, FastEthernet0/1
192.168.0.16/29
    [1] via 192.168.0.198, 00:00:01, FastEthernet0/1
192.168.0.24/29
    [1] via 192.168.0.190, 00:00:01, FastEthernet1/0
192.168.0.40/29    directly connected, FastEthernet2/0
192.168.0.184/29    directly connected, FastEthernet1/0
192.168.0.192/29    directly connected, FastEthernet0/1
192.168.0.200/29    directly connected, FastEthernet0/0
```

Configured DHCP for each router to assign every PC in the subnet a IP. (Use IP DHCP -r)

```
Pool 1 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.0.1          192.168.0.1      - 192.168.0.6       0
```

→ Excluding the router address!!.

Using traceroute command to see which router it hops to reach the destination and how many hops.

```
R4#traceroute 192.168.0.1

Type escape sequence to abort.
Tracing the route to 192.168.0.1

  1 192.168.0.206 76 msec 72 msec 72 msec
  2 192.168.0.1 2404 msec 108 msec 68 msec
```

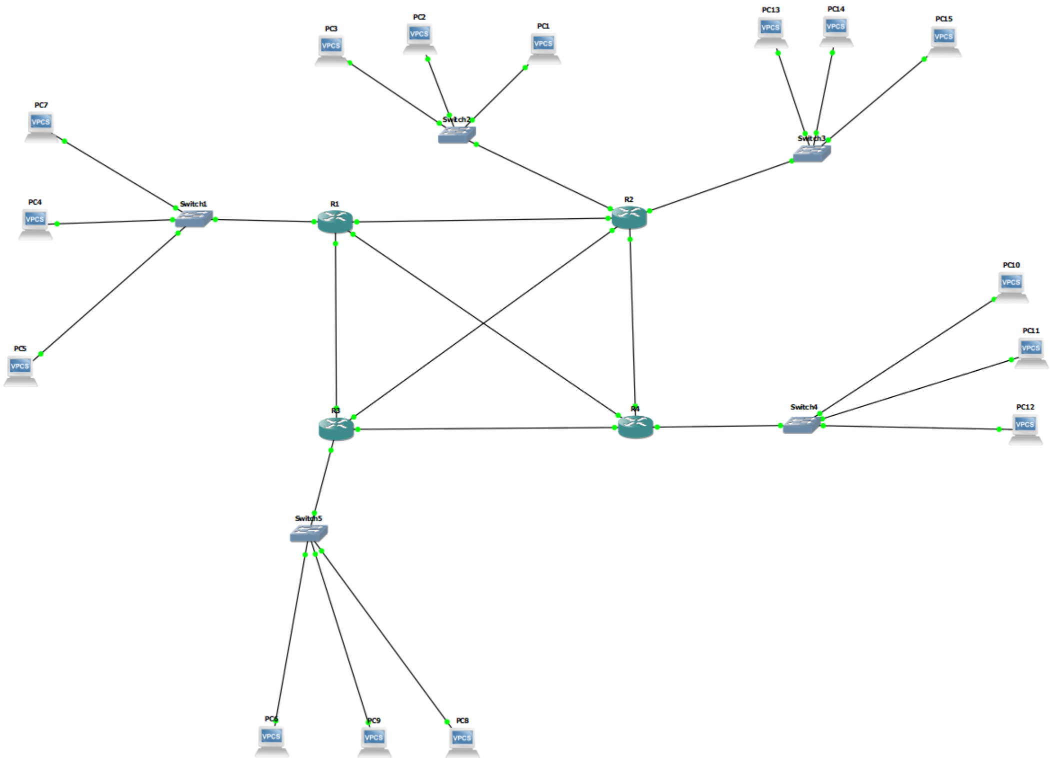
Pinging IP 102.168.0.45 from PC5 constantly to see what happens.

```
PC5> ping 192.168.0.41 -t
84 bytes from 192.168.0.41 icmp_seq=1 ttl=62 time=60.664 ms
84 bytes from 192.168.0.41 icmp_seq=2 ttl=62 time=59.896 ms
84 bytes from 192.168.0.41 icmp_seq=3 ttl=62 time=60.674 ms
84 bytes from 192.168.0.41 icmp_seq=4 ttl=62 time=46.123 ms
84 bytes from 192.168.0.41 icmp_seq=5 ttl=62 time=60.438 ms
84 bytes from 192.168.0.41 icmp_seq=6 ttl=62 time=59.498 ms
84 bytes from 192.168.0.41 icmp_seq=7 ttl=62 time=60.034 ms
84 bytes from 192.168.0.41 icmp_seq=8 ttl=62 time=60.855 ms
192.168.0.41 icmp_seq=9 timeout
192.168.0.41 icmp_seq=10 timeout
192.168.0.41 icmp_seq=11 timeout
192.168.0.41 icmp_seq=12 timeout
192.168.0.41 icmp_seq=13 timeout
192.168.0.41 icmp_seq=14 timeout
192.168.0.41 icmp_seq=15 timeout
192.168.0.41 icmp_seq=16 timeout
192.168.0.41 icmp_seq=17 timeout
192.168.0.41 icmp_seq=18 timeout
192.168.0.41 icmp_seq=19 timeout
```

→ The reason it stops receiving a reply back from the ping is because we stopped router 1 in the topology and since router 1 was the one navigating traffic between routers all the routers are disconnected from each other and hence considered a Single point of failure.

→ Before this TCP and UDP type pings also work fine!!!

<u>Part 2:-</u>



<u>Configuring IP addresses for each router interface with the corresponding subnet.</u> (Using different network ID but same subnet /29)

```
R1#sh ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.6.1.6        YES NVRAM  up                    up
FastEthernet0/1            10.6.1.206      YES NVRAM  up                    up
FastEthernet1/0            10.6.1.190      YES NVRAM  up                    up
FastEthernet2/0            10.6.1.198      YES NVRAM  up                    up
R1#
```

```
R2#sh ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.6.1.14       YES NVRAM  up                    up
FastEthernet0/1            10.6.1.205      YES NVRAM  up                    up
FastEthernet1/0            10.6.1.174      YES NVRAM  up                    up
FastEthernet2/0            10.6.1.182      YES NVRAM  up                    up
FastEthernet3/0            10.6.1.22       YES NVRAM  up                    up
```

```
R3#sh ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.6.1.38       YES NVRAM  up                    up
FastEthernet0/1            10.6.1.165      YES NVRAM  up                    up
FastEthernet1/0            10.6.1.189      YES NVRAM  up                    up
FastEthernet2/0            10.6.1.181      YES NVRAM  up                    up
R3#
```

```
R4#sh ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.6.1.30       YES NVRAM  up                    up
FastEthernet0/1            10.6.1.173      YES NVRAM  up                    up
FastEthernet1/0            10.6.1.166      YES NVRAM  up                    up
FastEthernet2/0            10.6.1.197      YES NVRAM  up                    up
R4#
```

Configuring RIP routing for each router based on connected IPs

```
R1#sh ip rip database
10.0.0.0/8    auto-summary
10.6.1.0/29    directly connected, FastEthernet0/0
10.6.1.8/29
    [1] via 10.6.1.205, 00:00:11, FastEthernet0/1
10.6.1.16/29
    [1] via 10.6.1.205, 00:00:11, FastEthernet0/1
10.6.1.24/29
    [1] via 10.6.1.197, 00:00:10, FastEthernet2/0
10.6.1.32/29
    [1] via 10.6.1.189, 00:00:04, FastEthernet1/0
10.6.1.160/29
    [1] via 10.6.1.197, 00:00:10, FastEthernet2/0
    [1] via 10.6.1.189, 00:00:04, FastEthernet1/0
10.6.1.168/29
    [1] via 10.6.1.197, 00:00:10, FastEthernet2/0
    [1] via 10.6.1.205, 00:00:11, FastEthernet0/1
10.6.1.176/29
    [1] via 10.6.1.205, 00:00:11, FastEthernet0/1
    [1] via 10.6.1.189, 00:00:04, FastEthernet1/0
10.6.1.184/29    directly connected, FastEthernet1/0
10.6.1.192/29    directly connected, FastEthernet2/0
10.6.1.200/29    directly connected, FastEthernet0/1
```

```
R2#sh ip rip database
10.0.0.0/8    auto-summary
10.6.1.0/29
    [1] via 10.6.1.206, 00:00:20, FastEthernet0/1
10.6.1.8/29    directly connected, FastEthernet0/0
10.6.1.16/29    directly connected, FastEthernet3/0
10.6.1.24/29
    [1] via 10.6.1.173, 00:00:19, FastEthernet1/0
10.6.1.32/29
    [1] via 10.6.1.181, 00:00:14, FastEthernet2/0
10.6.1.160/29
    [1] via 10.6.1.173, 00:00:19, FastEthernet1/0
    [1] via 10.6.1.181, 00:00:14, FastEthernet2/0
10.6.1.168/29    directly connected, FastEthernet1/0
10.6.1.176/29    directly connected, FastEthernet2/0
10.6.1.184/29
    [1] via 10.6.1.206, 00:00:20, FastEthernet0/1
    [1] via 10.6.1.181, 00:00:14, FastEthernet2/0
10.6.1.192/29
    [1] via 10.6.1.173, 00:00:19, FastEthernet1/0
    [1] via 10.6.1.206, 00:00:20, FastEthernet0/1
10.6.1.200/29    directly connected, FastEthernet0/1
```

```
R3#sh ip rip database
10.0.0.0/8    auto-summary
10.6.1.0/29
    [1] via 10.6.1.190, 00:00:21, FastEthernet1/0
10.6.1.8/29
    [1] via 10.6.1.182, 00:00:16, FastEthernet2/0
10.6.1.16/29
    [1] via 10.6.1.182, 00:00:16, FastEthernet2/0
10.6.1.24/29
    [1] via 10.6.1.166, 00:00:26, FastEthernet0/1
10.6.1.32/29    directly connected, FastEthernet0/0
10.6.1.160/29    directly connected, FastEthernet0/1
10.6.1.168/29
    [1] via 10.6.1.166, 00:00:26, FastEthernet0/1
    [1] via 10.6.1.182, 00:00:16, FastEthernet2/0
10.6.1.176/29    directly connected, FastEthernet2/0
10.6.1.184/29    directly connected, FastEthernet1/0
10.6.1.192/29
    [1] via 10.6.1.166, 00:00:26, FastEthernet0/1
    [1] via 10.6.1.190, 00:00:21, FastEthernet1/0
10.6.1.200/29
    [1] via 10.6.1.182, 00:00:16, FastEthernet2/0
    [1] via 10.6.1.190, 00:00:21, FastEthernet1/0
R3#
```

```
R4#sh ip rip database
10.0.0.0/8    auto-summary
10.6.1.0/29
    [1] via 10.6.1.198, 00:00:07, FastEthernet2/0
10.6.1.8/29
    [1] via 10.6.1.174, 00:00:19, FastEthernet0/1
10.6.1.16/29
    [1] via 10.6.1.174, 00:00:19, FastEthernet0/1
10.6.1.24/29    directly connected, FastEthernet0/0
10.6.1.32/29
    [1] via 10.6.1.165, 00:00:09, FastEthernet1/0
10.6.1.160/29    directly connected, FastEthernet1/0
10.6.1.168/29    directly connected, FastEthernet0/1
10.6.1.176/29
    [1] via 10.6.1.174, 00:00:19, FastEthernet0/1
    [1] via 10.6.1.165, 00:00:09, FastEthernet1/0
10.6.1.184/29
    [1] via 10.6.1.165, 00:00:09, FastEthernet1/0
    [1] via 10.6.1.198, 00:00:07, FastEthernet2/0
10.6.1.192/29    directly connected, FastEthernet2/0
10.6.1.200/29
    [1] via 10.6.1.174, 00:00:19, FastEthernet0/1
    [1] via 10.6.1.198, 00:00:07, FastEthernet2/0
R4#
```

Configured DHCP for each router to assign every PC in the subnet a IP. (Use IP DHCP -r)

```
R2#show ip dhcp pool

Pool 2 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.6.1.9             10.6.1.9         - 10.6.1.14          0

Pool 3 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 10.6.1.17            10.6.1.17        - 10.6.1.22          0
```

→ Excluding the router address!!.

Pinging IP 10.6.1.33 from PC3 constantly to see what happens.

```
PC3> ping 10.6.1.33 -t
84 bytes from 10.6.1.33 icmp_seq=1 ttl=62 time=61.809 ms
84 bytes from 10.6.1.33 icmp_seq=2 ttl=62 time=45.530 ms
84 bytes from 10.6.1.33 icmp_seq=3 ttl=62 time=60.063 ms
84 bytes from 10.6.1.33 icmp_seq=4 ttl=62 time=60.081 ms
84 bytes from 10.6.1.33 icmp_seq=5 ttl=62 time=60.637 ms
84 bytes from 10.6.1.33 icmp_seq=6 ttl=62 time=46.990 ms
84 bytes from 10.6.1.33 icmp_seq=7 ttl=62 time=60.060 ms
10.6.1.33 icmp_seq=8 timeout
10.6.1.33 icmp_seq=9 timeout
10.6.1.33 icmp_seq=10 timeout
10.6.1.33 icmp_seq=11 timeout
10.6.1.33 icmp_seq=12 timeout
10.6.1.33 icmp_seq=13 timeout
```

→ The ping stops because the router connected to that subnet turns off however other subnnets will still work and reply to the ping because the router are interconnected and hence defeats single point failure.

Pinging IP 10.6.1.25 from PC3 constantly when router connected to to IP 10.6.1.33 is OFF.

```
PC3> ping 10.6.1.25 -t
84 bytes from 10.6.1.25 icmp_seq=1 ttl=62 time=60.948 ms
84 bytes from 10.6.1.25 icmp_seq=2 ttl=62 time=61.813 ms
84 bytes from 10.6.1.25 icmp_seq=3 ttl=62 time=75.422 ms
84 bytes from 10.6.1.25 icmp_seq=4 ttl=62 time=61.934 ms
84 bytes from 10.6.1.25 icmp_seq=5 ttl=62 time=61.580 ms
84 bytes from 10.6.1.25 icmp_seq=6 ttl=62 time=61.282 ms
84 bytes from 10.6.1.25 icmp_seq=7 ttl=62 time=61.268 ms
84 bytes from 10.6.1.25 icmp_seq=8 ttl=62 time=60.706 ms
84 bytes from 10.6.1.25 icmp_seq=9 ttl=62 time=60.509 ms
84 bytes from 10.6.1.25 icmp_seq=10 ttl=62 time=60.916 ms
```

→ If we would want to make it in a way where the subnet will still respond when the corresponding gateway is off we would have to incoporate VRRP or virtual router redundancy protocol. This works by supplying a virtual router address to the PC to act as a gateway so incase one router (Master router) were to fail the backup router can work as a backup gateway and still reach the rest of the network even if the primary gateway is down.

→ Before this TCP and UDP type pings also work fine!!!

```
PC6> ping 10.6.1.17 -2
84 bytes from 10.6.1.17 udp_seq=1 ttl=62 time=61.215 ms
84 bytes from 10.6.1.17 udp_seq=2 ttl=62 time=60.648 ms
84 bytes from 10.6.1.17 udp_seq=3 ttl=62 time=61.711 ms
84 bytes from 10.6.1.17 udp_seq=4 ttl=62 time=61.173 ms
84 bytes from 10.6.1.17 udp_seq=5 ttl=62 time=60.611 ms

PC6> ping 10.6.1.17 -3
Connect   7@10.6.1.17 seq=1 ttl=62 time=74.657 ms
SendData  7@10.6.1.17 seq=1 ttl=62 time=75.153 ms
Close     7@10.6.1.17 seq=1 ttl=62 time=92.244 ms
Connect   7@10.6.1.17 seq=2 ttl=62 time=75.009 ms
SendData  7@10.6.1.17 seq=2 ttl=62 time=73.893 ms
Close     7@10.6.1.17 seq=2 ttl=62 time=90.505 ms
Connect   7@10.6.1.17 seq=3 ttl=62 time=76.047 ms
SendData  7@10.6.1.17 seq=3 ttl=62 time=75.510 ms
Close     7@10.6.1.17 seq=3 ttl=62 time=89.930 ms
Connect   7@10.6.1.17 seq=4 ttl=62 time=62.046 ms
SendData  7@10.6.1.17 seq=4 ttl=62 time=76.156 ms
Close     7@10.6.1.17 seq=4 ttl=62 time=90.403 ms
Connect   7@10.6.1.17 seq=5 ttl=62 time=61.180 ms
SendData  7@10.6.1.17 seq=5 ttl=62 time=75.578 ms
Close     7@10.6.1.17 seq=5 ttl=62 time=89.596 ms
```